Chapter 4: Clustering

Outline

4.1 Introduction

- Clustering is the unsupervised partitioning of data into natural groups.
- These groups are expected to be internally similar, externally different
- Clustering is useful for identifying operating regimes, that is, meaningful combinations of operating conditions.

Clustering methods

4.2 Clustering Methods

- Sequence to form clusters:
  - Flat - All clusters at the same time
  - Hierarchical - Sequentially
    - Agglomerative (bottom-up)
    - Divisive (top-down)
- Assumptions about data distribution
  - Centroid-based - Assume Gaussian distribution
  - Density-based - No assumptions about distribution

	Centroid-based	Density-based
Flat	K-means	DBSCAN
Hierarchical	Centroid linkage	HDBSCAN

4.2.1 Flat centroid-based clustering - K-means

- K-means seeks to minimize the sum of squared errors (SEE) within clusters.
- For each point in $\mathbf{X} = x_i, \ldots, x_n$ and each cluster in $C = c_k, \ldots, c_K$

$$SEE(C) = \sum_{k=1}^{K} \sum_{x_i \in c_k}^{n} \|x_i - \mu_k\|^2 \tag{1}$$

- Requires K preset, converges to local minima.
- Advantages: simple, efficient.
- Limitations: assumes spherical and equally sized clusters, sensitive to outliers.

4.2.1 Flat centroid-based clustering - K-means

Algorithm 1: K-Means Clustering

Input: K (number of clusters), dataset $X = \{x_1, \ldots, x_n\}$
Output: Centroids $C = \{c_1, \ldots, c_K\}$

Initialize centroids $C \leftarrow \{c_1, \ldots, c_K\}$ randomly;
repeat
    foreach *data point* $x \in X$ **do**
        Assign x to the closest centroid c_k;
    for $k \leftarrow 1$ **to** K **do**
        Update c_k as the mean of all points assigned to cluster k;
until *stopping criterion is met*;

4.2.2 Hierarchical density-based clustering

- Density-based clustering finds arbitrarily shaped clusters and detects noise.
- Hierarchical clustering finds arbitrarily sized clusters.

Density-based spatial clustering of applications with noise (DBSCAN)

$$N_{Eps}(p) = \{q \in D | dist(p, q) \leq Eps\} \quad (2)$$

- p is directly density-reachable (DDR) from q if $p \in N_{Eps}(q)$
- q is a core point if $N_{Eps} \geq MinPts$. A point is a border point if it is DDR from core points but core points are not DDR from it.
- Two points p, q are *density reachable* if there is a chain of core points connecting them. They are *density connected* if there is a third point o such that both p and q are density reachable from o.
- DBSCAN clusters are groups of density connected points

Hierarchical DBSCAN (HDBSCAN)

- Uses *mutual reachability distance* as a density metric.
- Builds a spanning tree from the mutual reachability distances to construct the hierarchy of the dataset.
- Better handles varying density scales; primary parameter often is minimum cluster size.

Unsupervised clustering metrics

4.3 Cluster evaluation metrics

- If clustering is an unsupervised task, how can we know if we got it right?
- We go back to the definition of *cluster* and use metrics to quantify the *goodness* of a clustering result based on how well it groups similar objects and how well it separates objects that are different.
- No single metric captures all the necessary qualities, so we use various metrics simultaneously and compare across iterations or configurations.
- Finally, we compare the clustering results against domain knowledge: does it make physical sense?

Silhouette score

- Measures cohesion vs separation per point; range [-1,1], higher is better.
- If C_i is the number of points in cluster i and d_{ij} is the distance between points i and j, the silhouette score can be computed as:

$$s_i = \begin{cases} 1 - a_i/b_i & \text{if } a_i < b_i, \\ 0 & \text{if } a_i = b_i, \\ b_i/a_i - 1 & \text{if } a_i > b_i \end{cases} \tag{3}$$

$$a_i = \frac{1}{|C_i| - 1} \sum_{i \neq j} d_{ij} \tag{4}$$

$$b_i = \min \left(\frac{1}{|C_i|} \right) \sum_{i \neq j} d_{ij} \tag{5}$$

Calinski-Harabasz, Davies-Bouldin, and S-Dbw index

- Calinski-Harabasz index
  - Ratio of the sum of the dispersion between clusters and of the sum of the dispersion withing clusters.
  - Assumes good clustering should have compact well separated clusters.
  - Higher is better but is unbounded.
- Davies-Bouldin index
  - Captures the average similarity between clusters.
  - Similarity, S_i, is the average distance from each point in a cluster to its centroid.
  - Lower (closer to 0) is better.
- S-Dbw index
  - Sum of average scattering for clusters and the inter-cluster density

S-Dbw index

- Combines intra-cluster scatter and inter-cluster density; lower indicates better clustering.
- Similar assumptions as Calinski-Harabasz.
- Lower is better.

Use of clustering metrics

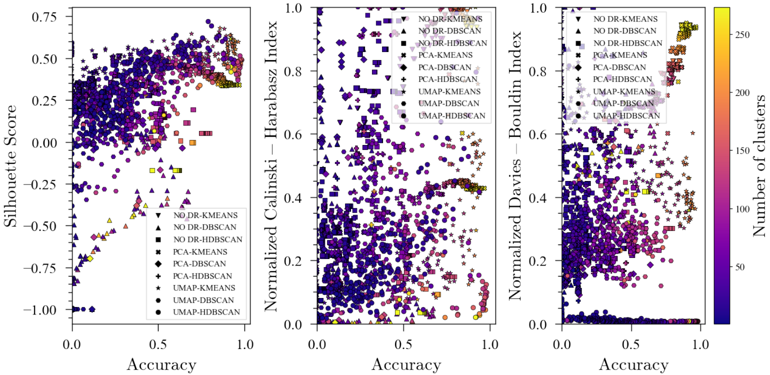

Figure: Clustering metrics for different combinations of DR and clustering methods. (Webb, Z. et al., Front. Chem. Eng., 2022)

- Clustering metrics rely on *distance metrics*.
- The most common distance metric is Euclidean:

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \tag{6}$$

- While Euclidean distance is good for lower dimensional spaces, higher dimensions of different data types might require different distance metrics.
- Some examples include Manhattan, cosine, and Jaccard distance.

4.4 Explaining Cluster Formation: Subspace Greedy Search (SGS)

- The goal is to identify which variables are the most influential in the cluster formation.
- Intuitively, we want to compare the distances across all dimensions for all points. Dimensions with larger distances are the most influential.
- However, this is computationally expensing. The number of comparisons for d dimensions is 2^d.
- SGS provides a computationally efficient way of doing this, by searching only a few subspaces through multiple iterations.
- Keeps high-scoring subspaces and prunes others to avoid combinatorial explosion.

4.4 SGS - Continued

Algorithm 2: Subspace Greedy Search (SGS)

Input: Variable list of all dimensions
Output: Ranked variables by frequency

```
Candidate_list ← [], Local_best ← [], Var_list ← all Vars;
d ← 1;
while d < length(Var_list) do
    Subspace_list ← all d-dimension combinations from Var_list
      that contain Candidate_list;
    Calculate Score for each Subspace in Subspace_list;
    Candidate_list ← Subspace with maximum score;
    Purge Var_list;
    Local_best.Add(Subspace with maximum score);
    d ← d + 1;
Rank variables in Local_best by frequency counting;
```

4.5 Hyperparameter tuning

- Selection of both DR and clustering methods can impact the result of the analysis.
- Additionally, each method may require the selection of user-defined parameters, such as number of neighbors in UMAP and minimum cluster size in HDBSCAN.
- These user-defined parameters are often called *hyperparameters*, and are important not only for unsupervised learning but for ML in general.
- The process of finding the right set of hyperparameters is called *hyperparameter tuning*.

4.5 Hyperparameter tuning - Continued

- Hyperparameter tuning is a search problem, as we need to find the best combination for our specific dataset.
- Some methods typically used include grid search and Bayesian optimization.
- Genetic algorithms are intuitive and effective methods that has shown good results in many applications.

4.5 Hyperparameter tuning - Continued

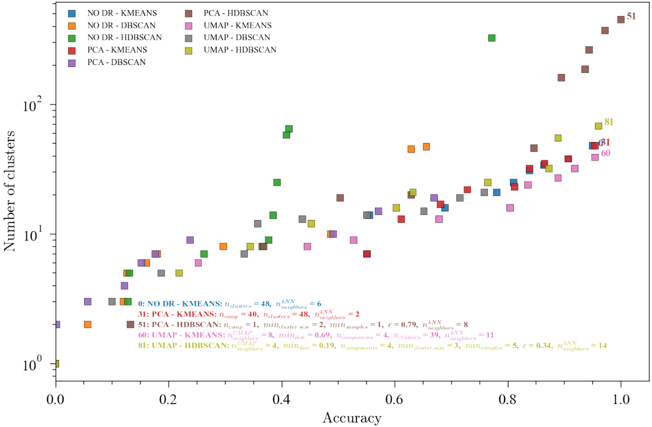

Figure: Best combinations of hyperparameters for DR and clustering ensembles using genetic algorithms (Webb, Z. et al., Front. Chem. Eng., 2022)

4.7 Conclusions

- Clustering is a powerful unsupervised learning technique for identifying patterns and groups in data.
- Different clustering methods (e.g., K-means, DBSCAN, HDBSCAN) have unique strengths and limitations, making method selection critical.
- Evaluation metrics like silhouette score, Calinski-Harabasz, and Davies-Bouldin help assess clustering quality but should be complemented with domain knowledge.
- Subspace Greedy Search (SGS) provides insights into the most influential variables driving cluster formation.
- Hyperparameter tuning is essential for optimizing clustering performance and requires careful consideration of the dataset and methods used.
- Clustering results should always be validated against domain knowledge to ensure meaningful and actionable insights.